Water Quality Index Modelling using Polynomial Regression

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INTRODUCTION

❖ What is WQI?

The Water Quality Index (WQI) is a mathematical tool used to transform complex water quality data into a single number. This number represents the overall quality of water at a certain location and time based on various physical, chemical, and biological parameters.

This Water Quality Index (WQI) model was developed using polynomial regression techniques.

OBJECTIVES

- To evaluate the overall quality of river water using the Water Quality Index (WQI).
- To assess the influence of key parameters (pH, DO, EC, and Nitrate) on water quality.
- To classify river water suitability based on standard drinking water guidelines.

PARAMETERS

➢ **pH (Potential of Hydrogen)**

pH indicates the acidic or alkaline nature of water and directly affects human health and aquatic life. Drinking water should have pH between 6.5–8.5; very low or high pH can cause health issues, bad taste, and harm fish reproduction.

➢ **DO (Dissolved Oxygen)**

Dissolved Oxygen (DO) indicates the amount of oxygen available in water and is essential for aquatic life.DO values of 4–6 mg/L indicate good water quality, while a rapid decrease in DO shows high organic pollution.

➢ **EC (Electrical Conductivity)**

Electrical Conductivity (EC) indicates the ability of water to conduct electricity, which depends on the presence of dissolved ions. Higher EC values indicate increased ionic concentration and possible pollution, while lower EC reflects better water quality.

➢ **Nitrate (NO··)**

Nitrate-Nitrogen (NO_3-N) indicates nutrient levels in water, mainly originating from agricultural runoff and sewage discharge. High nitrate-nitrogen concentrations degrade water quality and can cause eutrophication and health risks in drinking water.

DATA COLLECTION

- Water quality data for selected parameters were collected from government websites and published reports.
- These data show the actual measured values of water quality at selected locations.
- The standard values for comparison were taken from BIS IS-10500:2012 and other standard guidelines.

WEIGHTED ARITHMATIC WQI

The Water Quality Index (WQI) was calculated using the Weighted Arithmetic Index Method, which is widely used in India for water quality assessment. This method converts multiple water quality parameters into a single numerical value representing overall water quality.

➢ **Quality Rating Scale (Qi)**

For each parameter, the quality rating (Qi) was calculated using the following equation:

$$Qi = \frac{(Vi - V)}{(Si - V0)} \times 100$$

Vi = observed value
V0 = Ideal Value
Si = standard value

➢ **Unit Weight (Wi)**

The unit weight for each parameter was calculated using

$$Wi = \frac{k}{Si}$$

$k = \frac{1}{\sum Si}$

➢ **Final WQI Calculation**

The overall Water Quality Index was computed using:

$$WQI = \frac{\sum(Qi \times Wi)}{\sum Wi}$$

POLYNOMIAL REGRESSIONS

Polynomial regression analysis was used to study the relationship between water quality parameters and the calculated WQI values. This method helps in understanding non-linear trends and predicting variations in water quality.

Equation

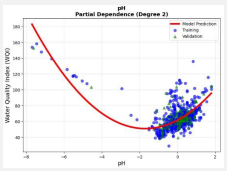
$$\hat{y} = a_0 + a_1(Dissolved\ Oxygen) + a_2(pH) + a_3(EC) + a_4(Nitrate)$$

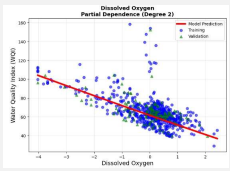
$\hat{y}$ = Water Quality Index (WQI)
x = water quality parameter
$a_0, a_1, a_2 \ldots\ldots a_n$ = coefficients

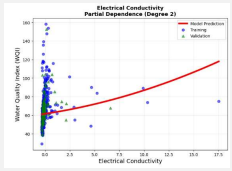
Parameter	Unit	Ideal Value	Standard / Permissible Value
pH	–	7.0	6.5–8.5 (IS 10500 / WHO)
Dissolved Oxygen (DO)	mg/L	14.6	≥ 5 (for good quality water)
Electrical Conductivity (EC)	µS/cm	0	≤ 1000 (drinking water)
Nitrate (NO_3^-)	mg/L	0	≤ 45 (IS 10500 / WHO)

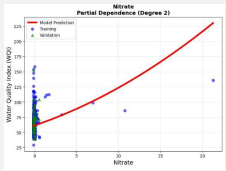
Value of WQI	Quality of Water
0–25	Excellent
26–50	Good
51–75	Poor
76–100	Very Poor
>100	Unsuitable for Drinking

SCATTER PLOT









REFERENCE

CPCB | Central Pollution Control Board, Ministry of Environment | Akkshay S. R. (Year). NWMP Water Quality Data for Indian Lakes. Kaggle.